

MODEL QUESTION PAPER 1

S6 INSTRUMENTATION ENGINEERING

COURSE ADVANCED BIOMEDICAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	What would cardiac pacemaker be classified as? a) a Therapeutic Instrument b) an Analytic Instrument c) a Diagnostic Instrument d) a Surgical Instrument	M1.02	R
2	Name any one device which is used as a Biomedical recorder.	M1.11	R
3	Where are microelectrodes applied?	M1.14	R
4	Ventricular fibrillation can be converted into a more efficient rhythm by applying a _____. Name two instruments used for nuclear radiation measurement.	M2.05	R
5	List the two types of ventilators.	M2.12	R
6	Mark the following statements as true or false: a) Cross-sectional images with any orientation are possible in NMR imaging systems b) NMR uses ionizing radiation	M3.12	R
7	Write the equation of Stefan Boltzman formula.	M3.18	R
8	State the purpose of cardiac monitor.	M4.02	R
9	Mention any two applications of telemedicine.	M4.10	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	List any three criteria for selecting biomedical transducers.	M1.16	R
2.	What are the differences between implantable pacemaker and external pacemaker?	M2.03	R
3.	State the need of defibrillators.	M2.05	R
4.	List the functions of haemodialysis machine.	M2.07	R
5.	What are the uses of respirators?	M2.09	R
6.	List the advantages and disadvantages of shortwave diathermy treatment.	M2.19	R
7.	List two limitations of conventional X-rays.	M3.07	R
8.	Compare thermal detectors and photo detectors.	M3.20	U
9.	State the objectives of patient monitoring system.	M4.01	R
10.	List three methods of testing biomedical equipments.	N4.20	R

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain the components of Man-Instrument system with the help of a block diagram. OR	M1.06	U
IV	Explain the problems encountered while performing Measurement of living organisms.	M1.09	U
V	Illustrate a microprocessor controlled ventilator and explain its operation. OR	M2.14	U
VI	Explain the working of electrical conductivity blood cell counter.	M2.23	U
VII	List the properties of X-rays. OR	M3.01	R
VIII	List the advantages of NMR imaging system.	M3.12	R
IX	Explain A-scan, M-scan and B-scan in an Ultrasonic imaging system. OR	M3.17	U
X	Explain the block diagram of IR imaging system.	M3.21	U
XI	Describe the working of a bedside monitor. OR	M4.03	U
XII	Explain the block diagram of biotelemetry system.	M4.07	U
XIII	Explain the effects of electric current on the human body. OR	M4.14	U
XIV	Explain the electrical safety analyzers used for biomedical instruments.	M4.19	U

Course: **ADVANCED BIOMEDICAL INSTRUMENTATION** – Question Paper 1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	1.02	R	1	2
I.2	1.11	R	1	2

I.3	1.14	R	1	2
I.4	2.05	R	1	2
I.5	2.12	R	1	2
I.6	3.12	R	1	2
I.7	3.18	R	1	2
I.8	4.02	R	1	2
I.9	4.10	R	1	2
II.1	1.16	R	3	8
II.2	2.03	R	3	8
II.3	2.05	R	3	8
II.4	2.07	R	3	8
II.5	2.09	R	3	8
II.6	2.19	R	3	8
II.7	3.07	R	3	8
II.8	3.20	U	3	8
II.9	4.01	R	3	8
II.10	4.20	R	3	8
III	1.06	U	7	16
IV	1.09	U	7	16
V	2.14	U	7	16
VI	2.23	U	7	16
VII	3.01	R	7	16
VIII	3.12	R	7	16
IX	3.17	U	7	16
X	3.21	U	7	16
XI	4.03	U	7	16
XII	4.07	U	7	16
XIII	4.14	U	7	16
XIV	4.19	U	7	16

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	13	21 ± 6	3	3	1	3	2	14	6	20
2	21	34.5 ± 6	2	2	5	15	2	14	9	31
3	20	33 ± 6	2	2	2	6	4	28	8	36
4	21	34.5 ± 6	2	2	2	6	4	28	8	36
Total	75	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	50	40.5
Understanding	73	59.5
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

(REVISION 2021)

MODEL QUESTION PAPER - 2

S6 INSTRUMENTATION ENGINEERING

SUBJECT : ADVANCED BIOMEDICAL INSTRUMENTATION

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	Give two examples for Therapeutic Instruments.	M1.02	R
2	The branch of science that includes the measurement of physiological variables and parameters is known as _____	M1.05	R
3	Name any two sources of biomedical signals.	M1.08	R
4	The process for removing impurity from blood is called _____	M2.07	R
5	List any two types of diathermy equipment.	M2.17	R
6	On computed tomography, bone is displayed as _____ c) White d) Black e) Grey	M3.08	R
7	Choose the correct pairing. A mode - Amplitude B mode - Bio-tissue M mode - Mobile	M3.17	R
8	Central stations are primarily designed for coronary care patients to display ECG waveforms and heart-rate information. How many patients can be monitored by Central station for this purpose? Define micro-shock.	M4.04	R
9	Define micro-shock.	M4.15	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	What is the role of Biomedical equipment technician in hospitals?	M1.04	R
2.	State the need of pacemakers.	M2.01	R
3.	What is a Defibrillator?	M2.05	R
4.	List the functions of a respirator.	M2.09	R
5.	State the need for anaesthesia.	M2.15	R
6.	What is the significance of blood cell counting?	M2.22	R
7.	List the physical factors which affect the amount of infrared radiation from the human body.	M3.19	R
8.	Explain the biological effects of NMR imaging.	M3.11	U
9.	List the functions of central station monitors.	M4.04	R
10.	Name any three essential parameters for telemedicine.	M4.11	R

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Describe the Block diagram of Biomedical Instrumentation system.	M1.01	U
	OR		
IV	Explain microelectrodes, skin surface electrodes and needle electrodes.	M1.14	U
V	Explain the working of DC defibrillator.	M2.06	U
	OR		
VI	Explain the working of Ultrasonic diathermy unit.	M2.21	U
VII	List the three techniques used for the visualization of X-rays. How X-rays are visualized using each of this technique?	M3.04	R
	OR		
VIII	List the properties of ultrasonic waves.	M3.14	R
IX	Explain the working of an X-ray machine with block diagram.	M3.03	U
	OR		
X	Explain the operation of CT scanner with block diagram.	M3.09	U
XI	Describe the applications of biotelemetry.	M4.08	U
	OR		
XII	Explain the principle of telemedicine using mobile communication with the help of a block diagram.	M4.13	U
XIII	Explain the following methods of accident prevention: i. Grounding ii. Double Insulation iii. Protection by Low Voltage	M4.18	U
	OR		U
XIV	Explain the three methods of testing biomedical equipments.	M4.20	

Question Wise Analysis

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.02	R	1	2
I.2	M1.05	R	1	2
I.3	M1.08	R	1	2
I.4	M2.07	R	1	2
I.5	M2.17	R	1	2
I.6	M3.08	R	1	2
I.7	M3.17	R	1	2
I.8	M4.04	R	1	2
I.9	M4.15	R	1	2
II.1	M1.04	R	3	8
II.2	M2.01	R	3	8
II.3	M2.05	R	3	8
II.4	M2.09	R	3	8
II.5	M2.15	R	3	8
II.6	M2.22	R	3	8
II.7	M3.19	R	3	8
II.8	M3.11	U	3	8
II.9	M4.04	R	3	8
II.10	M4.11	R	3	8
III	M1.01	U	7	16
IV	M1.14	U	7	16
V	M2.06	U	7	16
VI	M2.21	U	7	16
VII	M3.04	R	7	16
VIII	M3.14	R	7	16
IX	M3.03	U	7	16
X	M3.09	U	7	16
XI	M4.08	U	7	16
XII	M4.13	U	7	16
XIII	M4.18	U	7	16
XIV	M4.20	U	7	16

Course: **ADVANCED BIOMEDICAL INSTRUMENTATION** – Question Paper 2

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	50	40.5
Understanding	73	59.5
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100